Mutual Fund Analysis Report (2019–2024)

1. Project Overview and Objective

This project involves a comprehensive analysis of Mutual Fund data using **Excel** and **Power BI**. The primary goal is to evaluate fund performance, investor activity, and risk exposure to provide actionable insights for stakeholders.

- **Objective**: To clean, transform, and visualize mutual fund datasets to understand mobilization trends, redemption patterns, and the impact of risk and liquidity on Assets Under Management (AUM).

2. Data Source and Attributes

Dataset Type: Mutual Fund Historical Operational Data**.**

Dataset Source: [Indian Data Portal]

Domain: Finance / Investment Banking.

Key Attributes:

Fund Mobilized: New capital brought into the funds.

Repurchase/Redemption: Capital withdrawn by investors.

Net Inflow/Outflow: The net difference between mobilization and redemption.

AUM (Assets Under Management): Total market value of investments managed.

Folios: Individual investor accounts (Total vs. Segregated).

3. Problem Statement

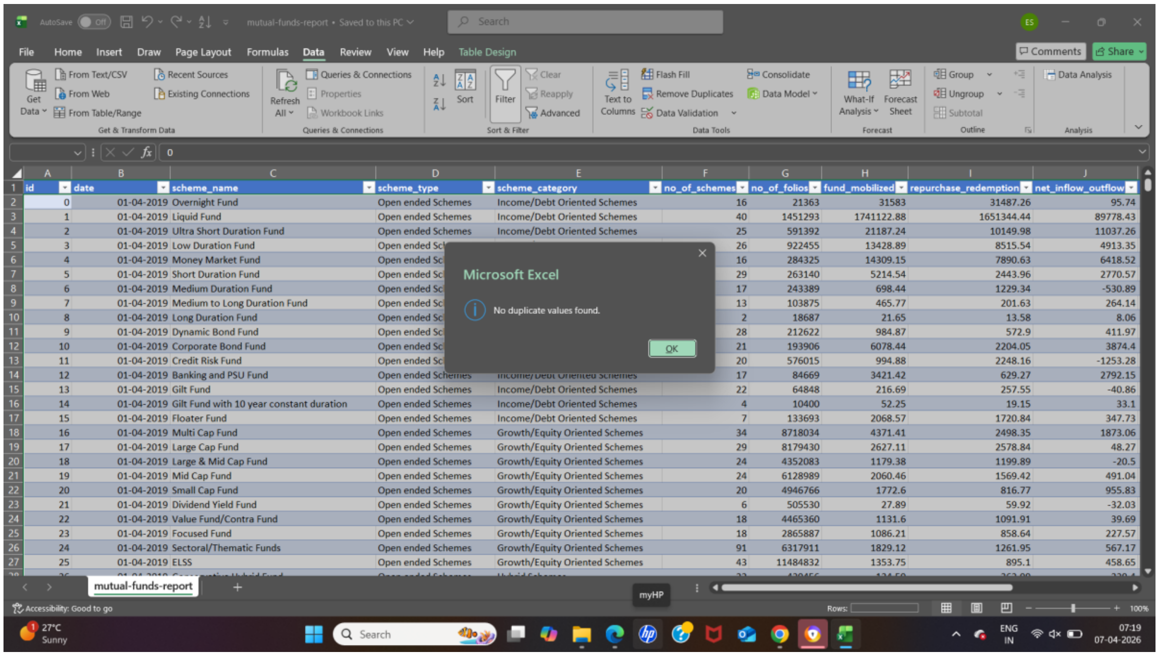
The analysis seeks to address the following critical business questions:

- What is the year-on-year trend for fund mobilization versus redemption?
- How do different scheme categories (Equity, Debt, Hybrid) contribute to the total AUM?
- What is the portfolio distribution across different risk levels?
- How does liquidity risk affect the overall asset management strategy?

4. Data Cleaning in Excel

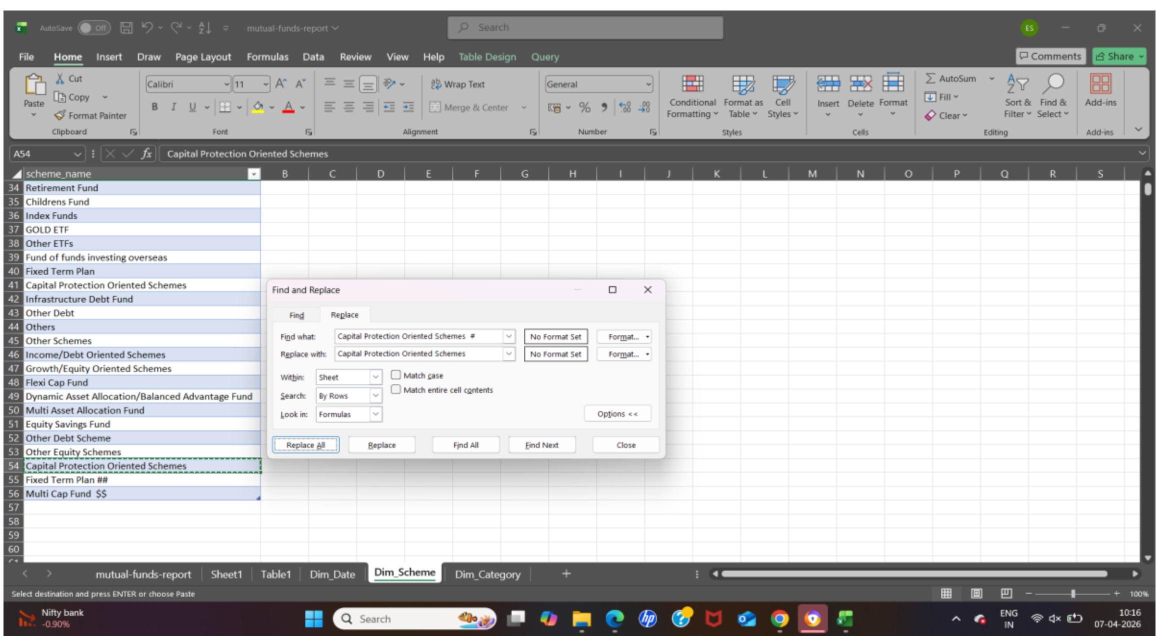
Step 1: Remove Duplicates

Used Excel Remove Duplicates option to ensure unique records.



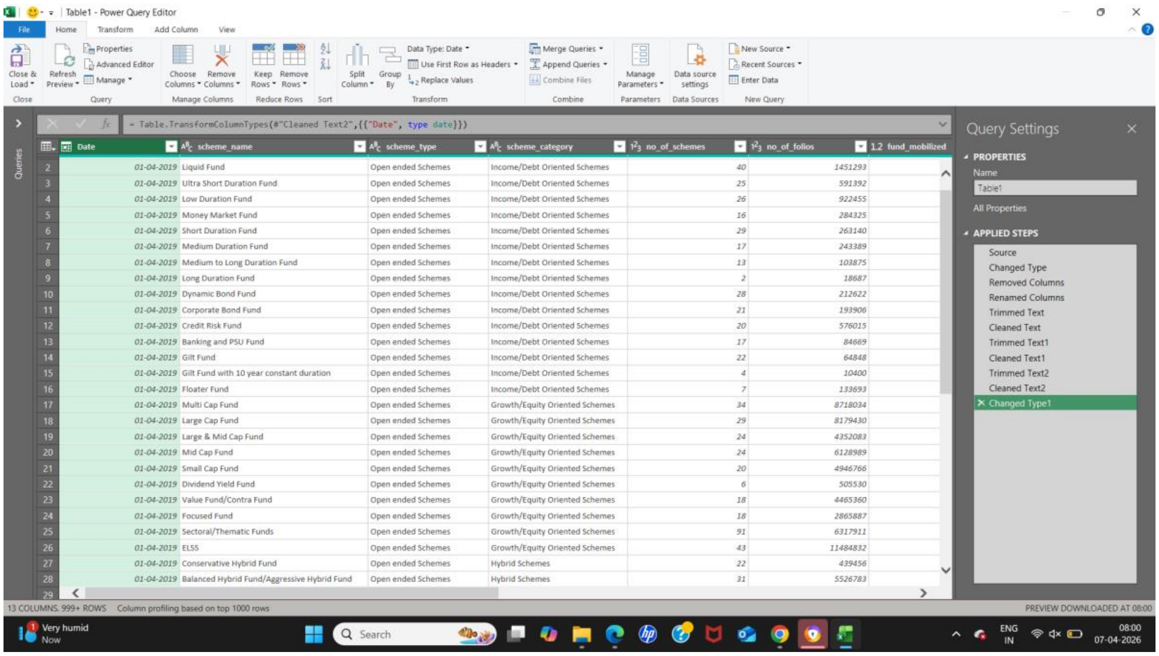
Step 2: Removing Unwanted Values

Checked for blanks and ensured numeric columns had valid values. Zero values were retained as they represent no activity.



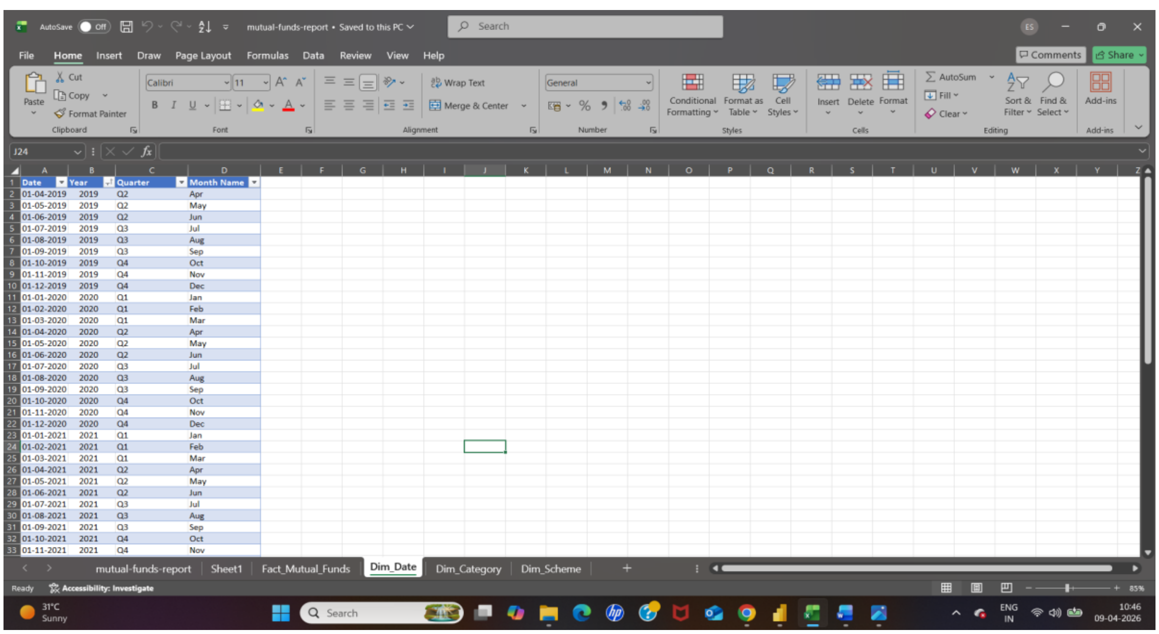
Step 3: Standardize Text

Used TRIM and CLEAN functions to remove extra spaces and inconsistencies.



Step 4: Create Date Columns

Extracted Year, Month, and Quarter using formulas:
=YEAR(Date)
=MONTH(Date), =TEXT([@Date],"mmm")
="Q"&ROUNDUP(MONTH(Date)/3,0)

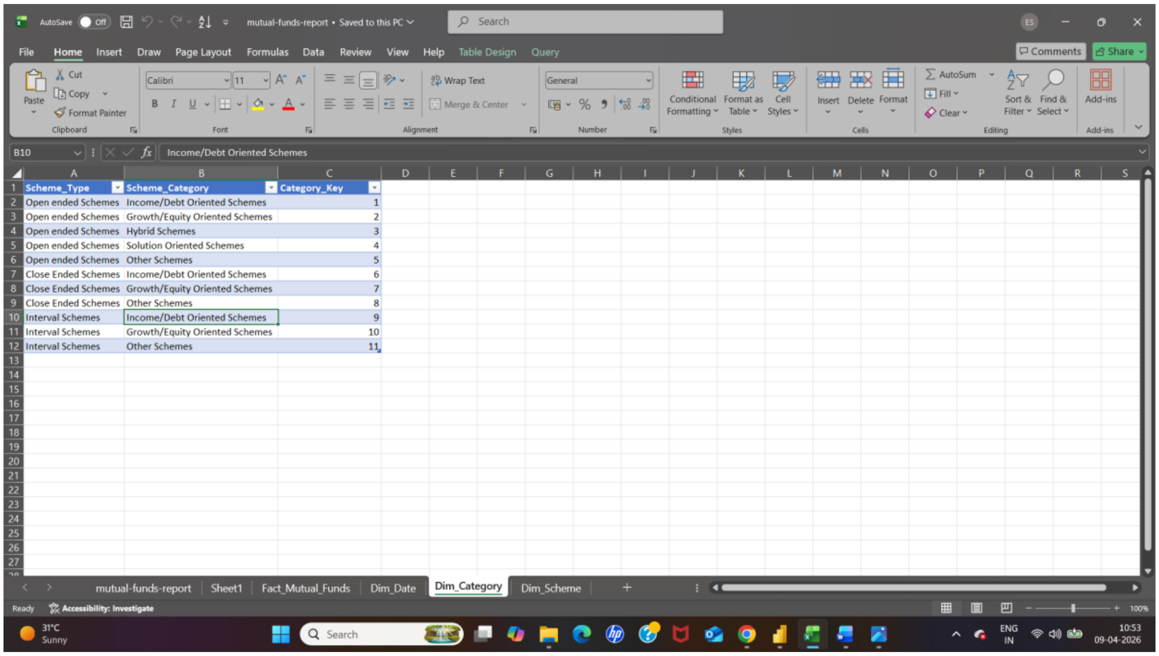


Step 5: Create Dimension Tables

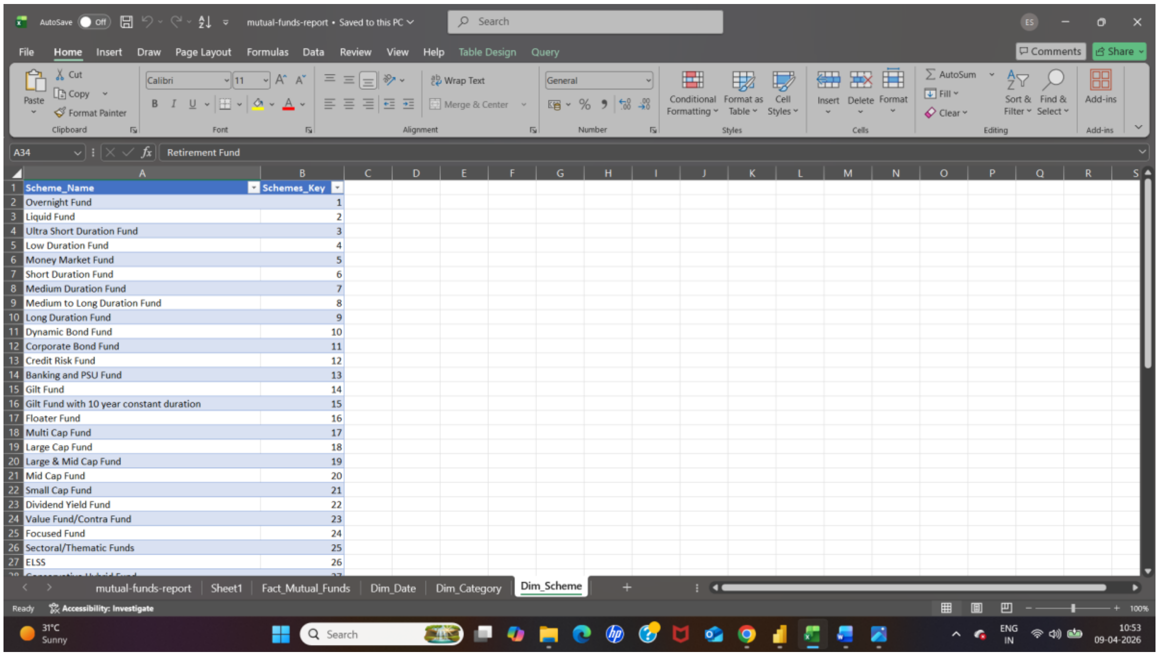
Created Dim_Date, Dim_Scheme, and Dim_Category tables using unique values.

Step 6: Create Keys

Assigned numeric surrogate keys (1,2,3...) for Scheme_Key and Category_Key.



Scheme_Type	Scheme_Category	Category_Key
Open ended Schemes	Income/Debt Oriented Schemes	1
Open ended Schemes	Growth/Equity Oriented Schemes	2
Open ended Schemes	Hybrid Schemes	3
Open ended Schemes	Solution Oriented Schemes	4
Open ended Schemes	Other Schemes	5
Close Ended Schemes	Income/Debt Oriented Schemes	6
Close Ended Schemes	Growth/Equity Oriented Schemes	7
Close Ended Schemes	Other Schemes	8
Interval Schemes	Income/Debt Oriented Schemes	9
Interval Schemes	Growth/Equity Oriented Schemes	10
Interval Schemes	Other Schemes	11



Scheme_Name	Schemes_Key
Overnight Fund	1
Liquid Fund	2
Ultra Short Duration Fund	3
Low Duration Fund	4
Money Market Fund	5
Short Duration Fund	6
Medium Duration Fund	7
Medium to Long Duration Fund	8
Long Duration Fund	9
Dynamic Bond Fund	10
Corporate Bond Fund	11
Credit Risk Fund	12
Banking and PSU Fund	13
Gilt Fund	14
Gilt Fund with 10 year constant duration	15
Floater Fund	16
Multi Cap Fund	17
Large Cap Fund	18
Large & Mid Cap Fund	19
Mid Cap Fund	20
Small Cap Fund	21
Dividend Yield Fund	22
Value Fund/Contra Fund	23
Focused Fund	24
Sectoral/Thematic Funds	25
ELSS	26

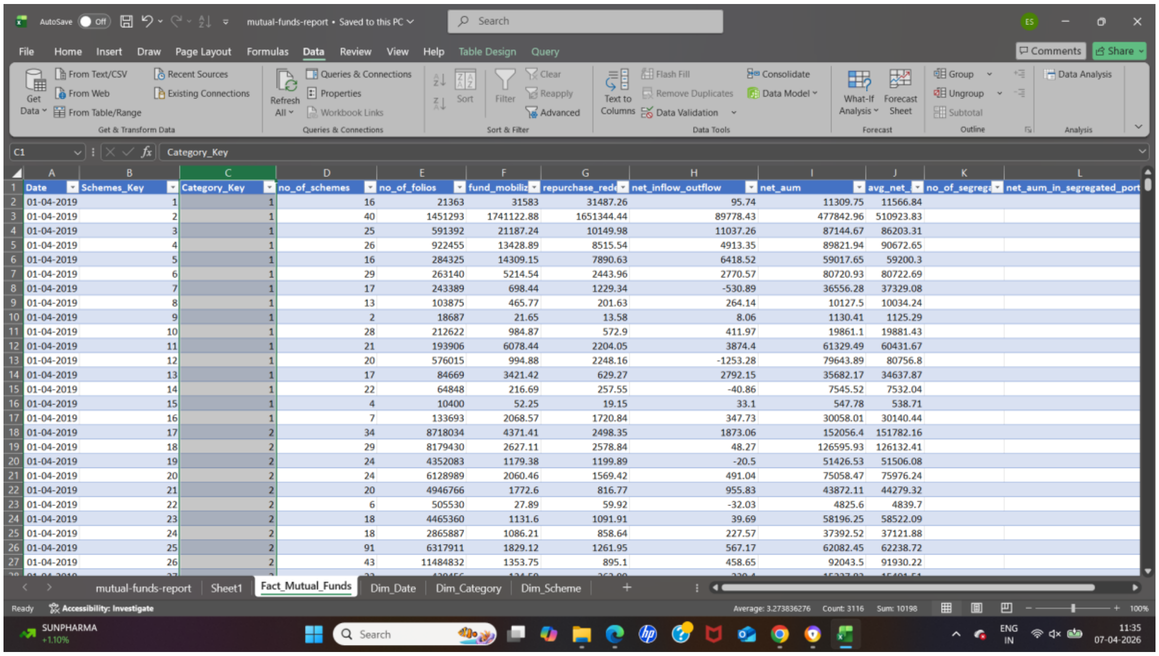
Step 7: Map Keys using XLOOKUP

Used XLOOKUP to replace text with keys in fact table.

Schemes_Key= XLOOKUP([@[scheme_name]],Table1_19[Scheme_Name],Table1_19[Schemes_Key])

Duplicate_Column= [@[scheme_type]]&"and"&[@[scheme_category]]

Category_Key= XLOOKUP([@Column1],Table1_18[Column1],Table1_18[Category_Key])



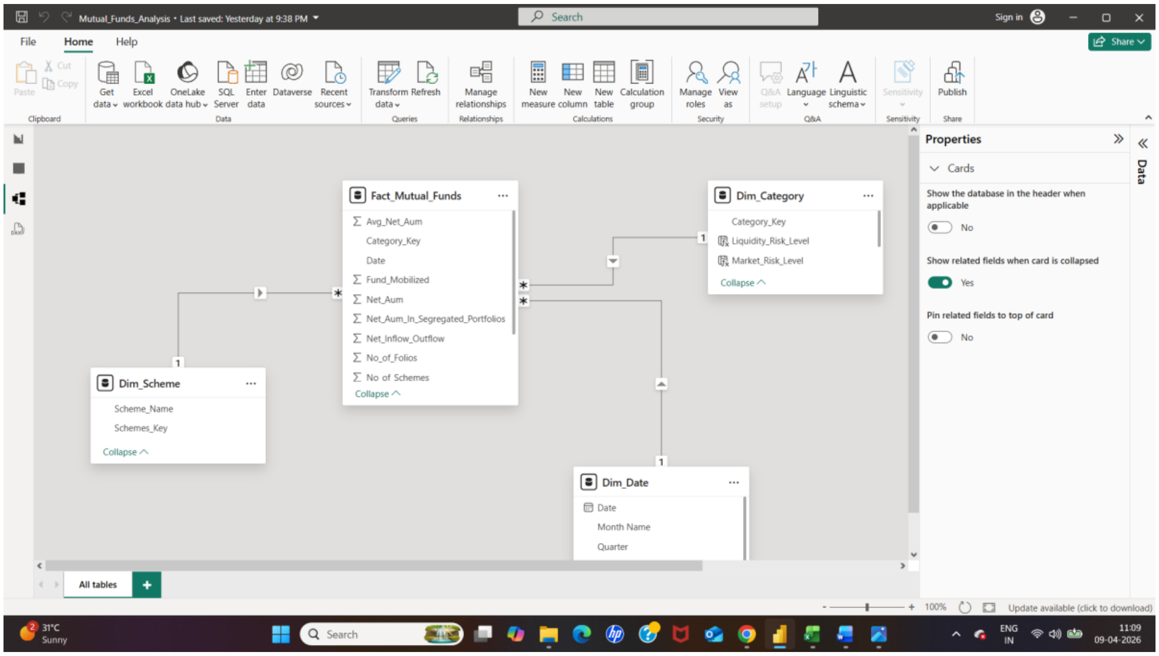
Date	Schemes_Key	Category_Key	no_of_schemes	no_of_folios	fund_mobiliz	repurchase_rede	net_inflow_outflow	net_aum	avg_net_	no_of_segrega	net_aum_in_segregated_port
01-04-2019	1	1	16	21363	31583	31487.26	95.74	11309.75	11566.84		
01-04-2019	2	1	40	1451293	1741122.88	1651344.44	89778.43	477842.96	510923.83		
01-04-2019	3	1	25	591392	21187.24	10149.98	11037.26	87144.67	86203.31		
01-04-2019	4	1	26	922455	13428.89	8515.54	4913.35	89821.94	90672.65		
01-04-2019	5	1	16	284325	14309.15	7890.63	6418.52	59017.65	59200.3		
01-04-2019	6	1	29	263140	5214.54	2443.96	2770.57	80720.93	80722.69		
01-04-2019	7	1	17	243389	698.44	1229.34	-530.89	36556.28	37329.08		
01-04-2019	8	1	13	103875	465.77	201.63	264.14	10127.5	10034.24		
01-04-2019	9	1	2	18687	21.65	13.58	8.06	1130.41	1125.29		
01-04-2019	10	1	28	212622	984.87	572.9	411.97	19861.1	19881.43		
01-04-2019	11	1	21	193906	6078.44	2204.05	3874.4	61329.49	60431.67		
01-04-2019	12	1	20	576015	994.88	2248.16	-1253.28	79643.89	80756.8		
01-04-2019	13	1	17	84669	3421.42	629.27	2792.15	35682.17	34637.87		
01-04-2019	14	1	22	64848	216.69	257.55	-40.86	7545.52	7532.04		
01-04-2019	15	1	4	10400	52.25	19.15	33.1	547.78	538.71		
01-04-2019	16	1	7	133693	2068.57	1720.84	347.73	30058.01	30140.44		
01-04-2019	17	2	34	8718034	4371.41	2498.35	1873.06	152056.4	151782.16		
01-04-2019	18	2	29	8179430	2627.11	2578.84	48.27	126595.93	126132.41		
01-04-2019	19	2	24	4352083	1179.38	1199.89	-20.5	51426.53	51506.08		
01-04-2019	20	2	24	6128989	2060.46	1569.42	491.04	75058.47	75976.24		
01-04-2019	21	2	20	4946766	1772.6	816.77	955.83	43872.11	44279.32		
01-04-2019	22	2	6	505530	27.89	59.92	-32.03	4825.6	4839.7		
01-04-2019	23	2	18	4465360	1131.6	1091.91	39.69	58196.25	58522.09		
01-04-2019	24	2	18	2865887	1086.21	858.64	227.57	37392.52	37121.88		
01-04-2019	25	2	91	6317911	1829.12	1261.95	567.17	62082.45	62238.72		
01-04-2019	26	2	43	11484832	1353.75	895.1	458.65	92043.5	91930.22		

5. Data Modeling

Created a star schema with Fact table and dimension tables connected using keys.

A **Star Schema** was implemented to optimize performance and reporting:

- **Fact Table:** Fact_Mutual_Funds (containing transactional data like AUM and flows).
- **Dimension Tables:** Dim_Date, Dim_Scheme, and Dim_Category



6. DAX Calculations

Market Risk Level

Categorized schemes into risk levels based on scheme category.

```
Market_Risk_Level = SWITCH(
    TRUE(),
    Dim_Category[Scheme_Category]="Income/Debt Oriented Schemes","Low Risk",
    Dim_Category[Scheme_Category]="Growth/Equity Oriented Schemes","High Risk",
    Dim_Category[Scheme_Category]="Hybrid Schemes","Moderate Risk",
    Dim_Category[Scheme_Category]="Solution Oriented Schemes","Moderate To High Risk",
    "Others"
)
```

Liquidity Risk Level

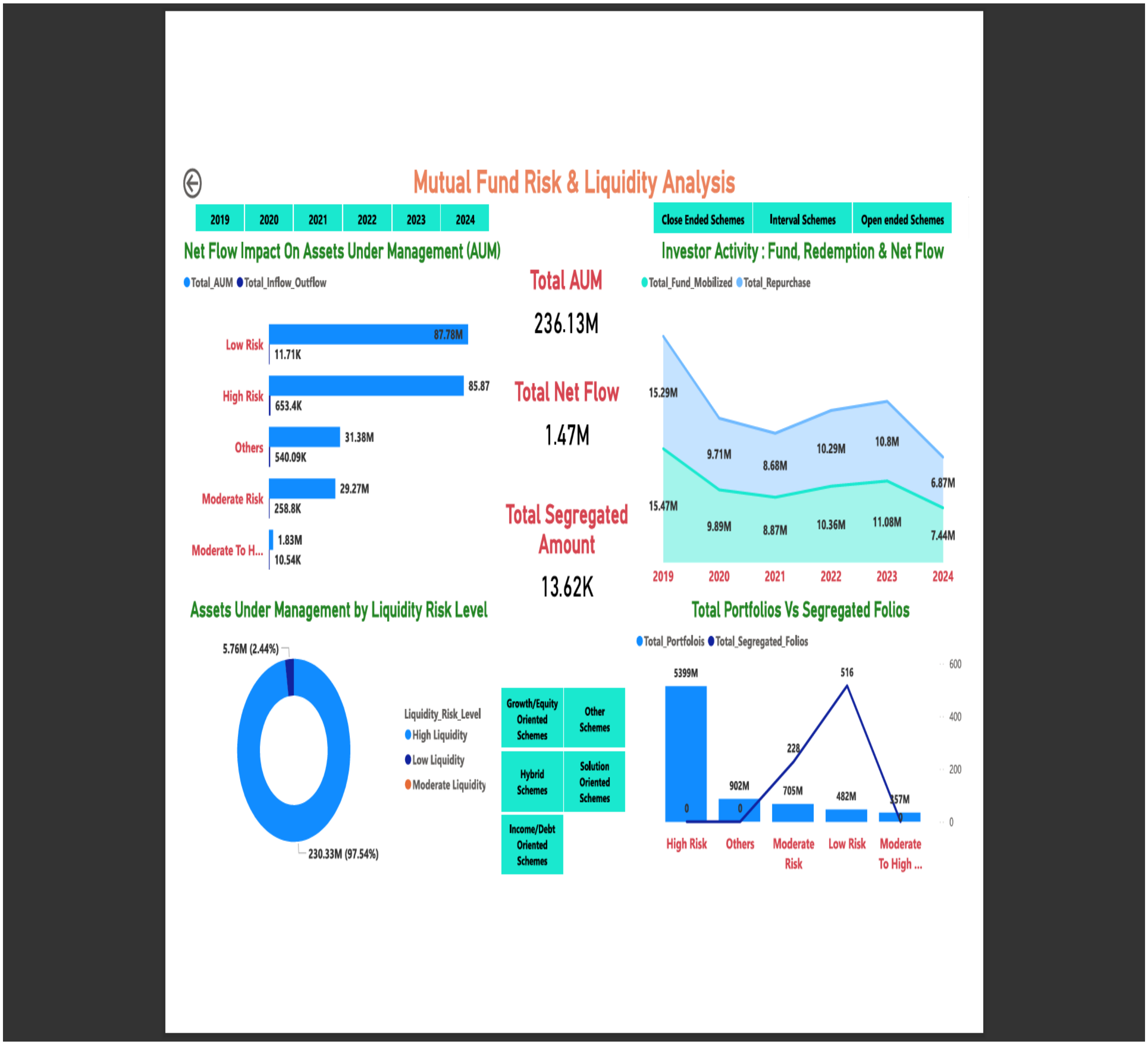
Categorized schemes based on liquidity using scheme type.

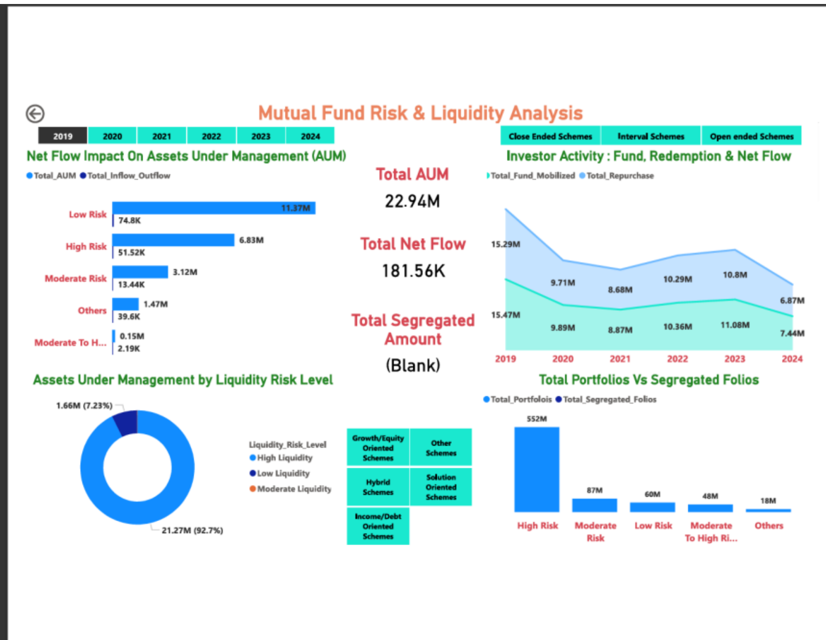
```
Liquidity_Risk_Level = SWITCH(
    TRUE(),
    Dim_Category[Scheme_Type]="Open ended Schemes","High Liquidity",
    Dim_Category[Scheme_Type]="Close Ended Schemes","Low Liquidity",
    "Moderate Liquidity"
)
```

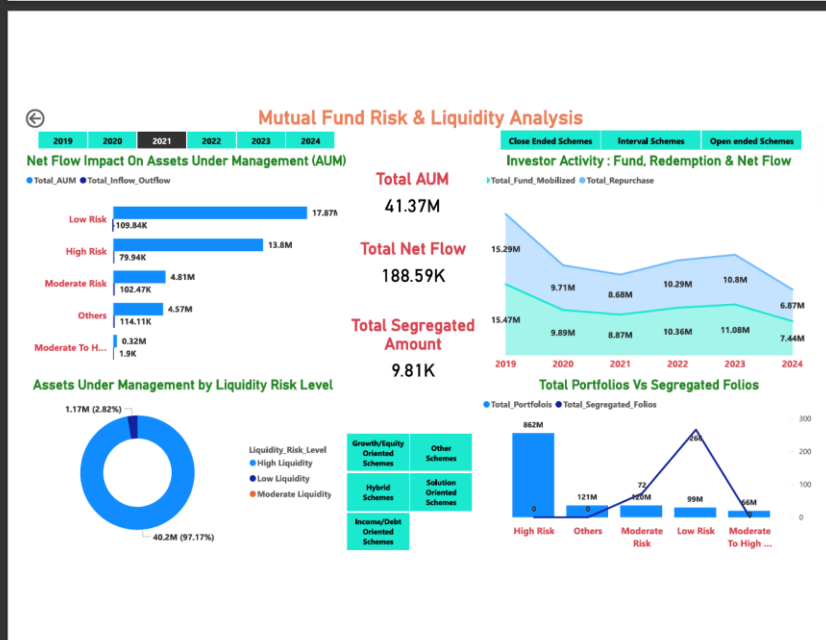
Other Dax Measures

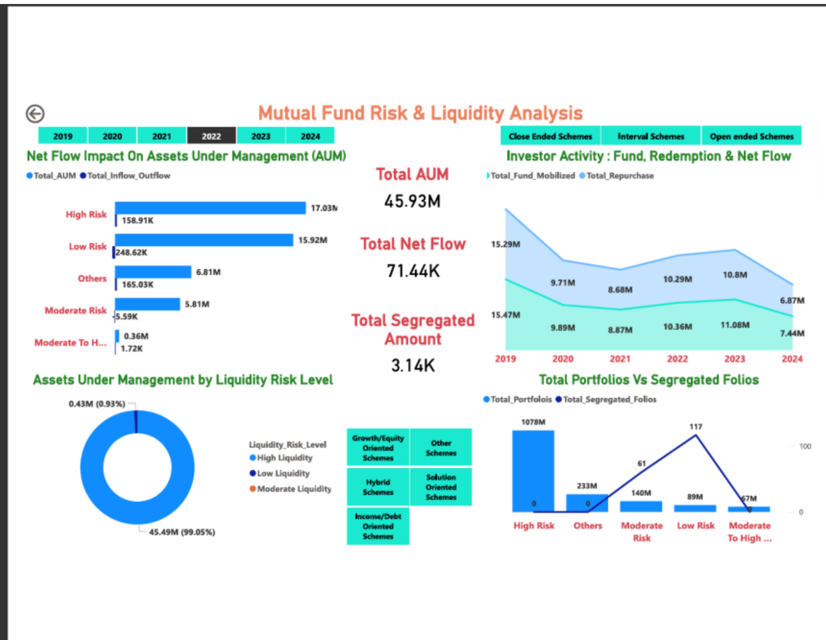
1.Total_AUM = SUM(Fact_Mutual_Funds[Net_Aum])
2.Total_Fund_Mobilized = SUM(Fact_Mutual_Funds[Fund_Mobilized])
3.Total_Inflow_Outflow = SUM(Fact_Mutual_Funds[Net_Inflow_Outflow])
4.Total_Portfolois = SUM(Fact_Mutual_Funds[No_of_Folios])
5.Total_Repurchase = SUM(Fact_Mutual_Funds[Repurchase_Redemption])
6.Total_Segregated_Folios = SUM(Fact_Mutual_Funds[No_Of_Segregated_Portfolios])
7.Total_Segregatted_Amount = SUM(Fact_Mutual_Funds[Net_Aum_In_Segregated_Portfolios])

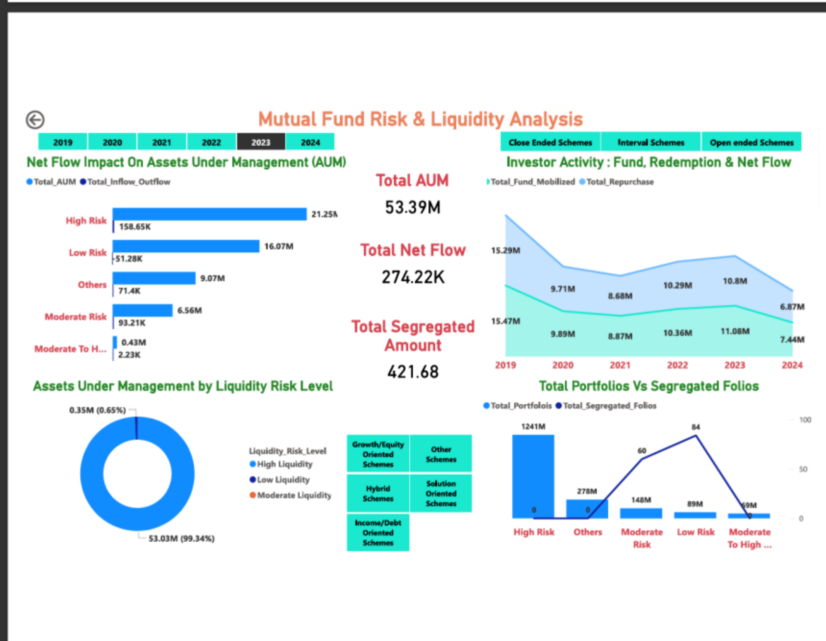
7. Charts and Dashboards

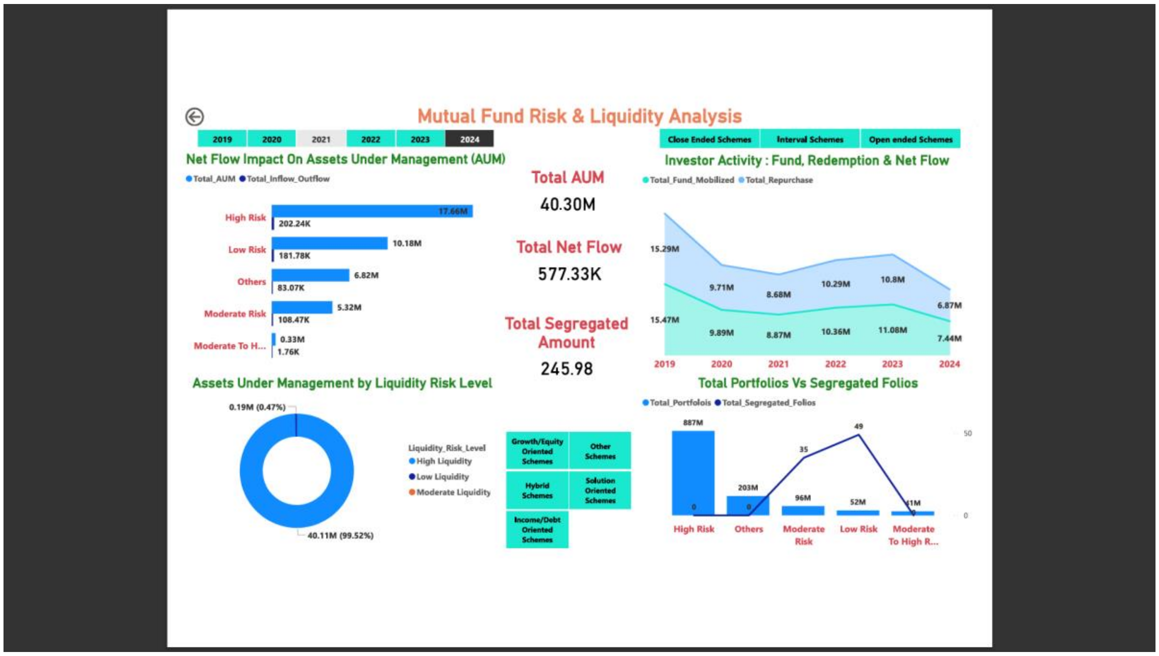



Mutual Fund Risk & Liquidity Analysis
2019
2020
2021
2022
2023
2024
Close Ended Schemes
Interval Schemes
Open ended Schemes
Net Flow Impact On Assets Under Management (AUM)
Total_AUM
Total_Inflow_Outflow
Low Risk
11.37M
74.8K
High Risk
6.83M
51.52K
Moderate Risk
3.12M
13.44K
Others
1.47M
39.6K
Moderate To H...
0.15M
2.19K
Total AUM
22.94M
Total Net Flow
181.56K
Total Segregated Amount
(Blank)
Investor Activity : Fund, Redemption & Net Flow
Total_Fund_Mobilized
Total_Repurchase
15.29M
9.71M
8.68M
10.29M
10.8M
6.87M
15.47M
9.89M
8.87M
10.36M
11.08M
7.44M
2019
2020
2021
2022
2023
2024
Assets Under Management by Liquidity Risk Level
1.66M (7.23%)
21.27M (92.7%)
Liquidity_Risk_Level
High Liquidity
Low Liquidity
Moderate Liquidity
Growth/Equity Oriented Schemes
Other Schemes
Hybrid Schemes
Solution Oriented Schemes
Income/Debt Oriented Schemes
Total Portfolios Vs Segregated Folios
Total_Portfolois
Total_Segregated_Folios
552M
87M
60M
48M
18M
High Risk
Moderate Risk
Low Risk
Moderate To High Ri...
Others


Mutual Fund Risk & Liquidity Analysis
2019
2020
2021
2022
2023
2024
Close Ended Schemes
Interval Schemes
Open ended Schemes
Net Flow Impact On Assets Under Management (AUM)
Total_AUM
Total_Inflow_Outflow
Low Risk
17.87M
109.84K
High Risk
13.8M
79.94K
Moderate Risk
4.81M
102.47K
Others
4.57M
114.11K
Moderate To H...
0.32M
1.9K
Total AUM
41.37M
Total Net Flow
188.59K
Total Segregated Amount
9.81K
Investor Activity : Fund, Redemption & Net Flow
Total_Fund_Mobilized
Total_Repurchase
15.29M
9.71M
8.68M
10.29M
10.8M
6.87M
15.47M
9.89M
8.87M
10.36M
11.08M
7.44M
2019
2020
2021
2022
2023
2024
Assets Under Management by Liquidity Risk Level
1.17M (2.82%)
40.2M (97.17%)
Liquidity_Risk_Level
High Liquidity
Low Liquidity
Moderate Liquidity
Growth/Equity Oriented Schemes
Other Schemes
Hybrid Schemes
Solution Oriented Schemes
Income/Debt Oriented Schemes
Total Portfolios Vs Segregated Folios
Total_Portfolois
Total_Segregated_Folios
862M
121M
120M
99M
66M
0
72
300
200
100
High Risk
Others
Moderate Risk
Low Risk
Moderate To High ...


Mutual Fund Risk & Liquidity Analysis
2019
2020
2021
2022
2023
2024
Close Ended Schemes
Interval Schemes
Open ended Schemes
Net Flow Impact On Assets Under Management (AUM)
Total_AUM
Total_Inflow_Outflow
High Risk
Low Risk
Others
Moderate Risk
Moderate To H...
17.03M
158.91K
15.92M
248.62K
6.81M
165.03K
5.81M
5.59K
0.36M
1.72K
Total AUM
45.93M
Total Net Flow
71.44K
Total Segregated Amount
3.14K
Investor Activity : Fund, Redemption & Net Flow
Total_Fund_Mobilized
Total_Repurchase
15.29M
9.71M
8.68M
10.29M
10.8M
6.87M
15.47M
9.89M
8.87M
10.36M
11.08M
7.44M
Assets Under Management by Liquidity Risk Level
0.43M (0.93%)
45.49M (99.05%)
Liquidity_Risk_Level
High Liquidity
Low Liquidity
Moderate Liquidity
Growth/Equity Oriented Schemes
Other Schemes
Hybrid Schemes
Solution Oriented Schemes
Income/Debt Oriented Schemes
Total Portfolios Vs Segregated Folios
Total_Portfolois
Total_Segregated_Folios
1078M
233M
140M
89M
67M
117
61
100
0
Moderate Risk
Moderate To High ...


Mutual Fund Risk & Liquidity Analysis
2019
2020
2021
2022
2023
2024
Close Ended Schemes
Interval Schemes
Open ended Schemes
Net Flow Impact On Assets Under Management (AUM)
Total_AUM
Total_Inflow_Outflow
High Risk
Low Risk
Others
Moderate Risk
Moderate To H...
21.25M
158.65K
16.07M
51.28K
9.07M
71.4K
6.56M
93.21K
0.43M
2.23K
Total AUM
53.39M
Total Net Flow
274.22K
Total Segregated Amount
421.68
Investor Activity : Fund, Redemption & Net Flow
Total_Fund_Mobilized
Total_Repurchase
15.29M
9.71M
8.68M
10.29M
10.8M
6.87M
15.47M
9.89M
8.87M
10.36M
11.08M
7.44M
Assets Under Management by Liquidity Risk Level
0.35M (0.65%)
53.03M (99.34%)
Liquidity_Risk_Level
High Liquidity
Low Liquidity
Moderate Liquidity
Growth/Equity Oriented Schemes
Other Schemes
Hybrid Schemes
Solution Oriented Schemes
Income/Debt Oriented Schemes
Total Portfolios Vs Segregated Folios
Total_Portfolois
Total_Segregated_Folios
1241M
278M
148M
89M
69M
84
60
100
50
0
Moderate Risk
Moderate To High ...



8.Observations and Insights

1. Year-on-Year Portfolio & Activity Comparison

The Portfolio Explosion (Users):

2019–2021: Growth was steady and focused on traditional safety.

2023–2024: We saw a massive jump in **High Risk (Equity**) portfolios, now totalling **5,399M** accounts. This proves the **"Retail Revolution"** is here—more individual people are investing in high-risk categories than ever before.

The Investment Cycle (Fund Mobilization):

2019 (The Benchmark): Peak investment year with **15.5M** coming in.

2023 (The Recovery): A strong comeback with **11.1M** as markets stabilized post-pandemic.

2024 (The Cautious Phase): Currently at **7.4M.** While lower, the quality of investors is higher, focusing on specific equity growth.

The Liquidity Shift: Over the years, investors have moved away from **"Close-ended"** portfolios. Today, **97.54%** of all portfolios are **Open-Ended**, showing that modern investors demand the ability to move their money instantly every year.

2. Descriptive Analysis (The Current Money Map)

AUM Breakdown: Our total Assets Under Management (AUM) is **236.13M**.

Low Risk (Debt): Holds the largest "store of value" at **87.78M.**

High Risk (Equity): Close second at **85.87M**, but with the highest number of active accounts **(5,399M).**

The 2019 Benchmark: In 2019, we saw peak excitement with **15.5M** in new capital. 2024's current level of **7.4M** suggests the market is **"coiling"**—investors are holding onto cash, waiting for a clear signal to jump back in.

Portfolio Health: We track **Total vs. Segregated Folios**. While Equity has zero segregated (stuck) folios, Debt has **516**, which is a key metric for risk-averse investors to watch.

Liquidity dominance: **97.54% of AUM** is in High **Liquidity schemes**. This shows that today's investors value "Exit Strategy" as much as "Entry Strategy."

3. Diagnostic Analysis (Why Investors are Moving Money)

The Volatility Play: High-risk investors are actually thriving on volatility. The **653.4K** Net Inflow in **High Risk (Equity)** shows that when the market moves, active investors see it as a buying opportunity.

The "Safety" Paradox: Even though **Low Risk (Debt)** has the most money **(87.78M)**, it has the lowest net growth **(11.71K)**. Large institutional investors use Debt as a **"Parking Lot"** for cash, but aren't adding new money due to credit concerns (Segregated Folios).

The Redemption Trigger: In **2023**, we saw **11.1M** come in and **10.8M** go out. This **"high-churn"** behaviour suggests investors are becoming more tactical—taking small profits quickly.

Niche Markets: **Solution-Oriented Schemes (Moderate to High Risk)** have tiny AUM **(1.83M)** because they lack the flexibility **(liquidity)** that modern investors demand.

4. Predictive Analysis (Forecasting Market Behaviour)

The Great Rotation: Based on account growth, we project that **High Risk (Equity)** AUM will **surpass** Low Risk (Debt) AUM by **mid-2025**. The "retail army" is moving into equity faster than institutions are moving into debt.

Volatility Forecast: As long as **97%+** of the fund remains in **"Withdraw Anytime"** status, we should expect sudden shifts in AUM. A **5%** market dip could trigger a **10%** withdrawal spike due to ease of access.

The Search for Yield: Because Low Risk returns are currently flat, we predict a massive shift toward **Hybrid Schemes (Moderate Risk).** They currently have **258.8K** in Net Inflows, proving they are the preferred **"Safe Haven"** for **2025**.

Segregated Recovery: If we clear the **421.68** in Segregated Amounts, we predict a **20-25%** surge in new Debt investments as trust is restored.

5. Prescriptive Analysis (The Investor's Action Plan)

Strategic Allocation: For high returns, follow the crowd into **High Risk (Equity)**—this is where the liquidity and net inflows are strongest.

Stability Strategy: For capital preservation, the **87.78M AUM** in Debt remains the safest anchor, provided you monitor the **"Segregated Folio"** count.

The "Middle Way": We recommend increasing exposure to **Hybrid Schemes**. It is currently the most balanced category, showing steady growth **(29.27M AUM)** without the extreme "In-and-Out" volatility of Equity.

Cash Reserves: The fund must maintain a **10-15%** cash buffer to handle the high volume of Outflows, ensuring no investor is ever denied a withdrawal during **high market volatility.**

Modernization: To return to 2019 investment levels, we must simplify **"Solution-Oriented"** products to make them as attractive and liquid as the Equity schemes.

9.Conclusion

This project provides a comprehensive, four-tier analysis of a 236.13M Mutual Fund portfolio from 2019 to 2024, revealing a massive structural shift where Growth/Equity (High Risk) has exploded to 5,399M portfolios and is projected to overtake Income/Debt (Low Risk) as the primary AUM driver by 2025. While 2019 marked the historical peak for fund mobilization at 15.5M, the current 2024 landscape of 7.4M reflects a tactical "wait and watch" approach by investors who are increasingly prioritizing the 97.54% high-liquidity open-ended schemes to navigate market volatility. Diagnostic insights highlight a "Safety Paradox" where Debt holds the most wealth (87.78M) but suffers from stagnant net inflows (11.71K) due to credit concerns and segregated folios, whereas Equity thrives with a dominant 653.4K net inflow despite higher risks. Ultimately, the report prescribes a strategic roadmap that balances aggressive equity marketing with a fix for debt-market trust and the promotion of Hybrid schemes (Moderate Risk) as the ideal "middle ground" safe haven for 2025, while maintaining high cash reserves to manage frequent redemption cycles.